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Directions for Use

B. Braun Mannitol 20%

Solution for Infusion

1. NAME OF THE MEDICINAL PRODUCT

B. Braun Mannitol 20% Solution for Infusion

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

1000 ml of solution contain

Mannitol	200.0 g

For a full list of excipients, see section 6.1

3. PHARMACEUTICAL FORM

Solution for infusion;
Product description: Clear, colourless aqueous solution

Physico-chemical characteristics:

Theoretical osmolarity	1100 mOsm/l
Titration acidity (to pH 7.4)	< 0.2 mmol/l
pH	4.5 - 7.0

4. CLINICAL PARTICULARS

4.1 Therapeutic Indications

- Prevention of acute renal failure (after positive response to test infusion);
- Reduction of intracranial pressure,
- Forced diuresis to promote the urinary excretion of toxic substances;
- Supportive systemic therapy of acute glaucoma

4.2 Posology and Method of Administration

Recommended dosages

Dosage and frequency of intravenous infusions of the mannitol solutions are determined according to the prevailing indication, clinical and biological condition of the patient, weight and concomitant therapy.

The lowest dose producing the desired effect should be applied.

Adults and the elderly

As a general rule 25 - 100 g of mannitol are administered per day, corresponding to (approx.) 125 – 500 ml of B. Braun Mannitol 20% Solution for Infusion. Larger doses up to 200 g of mannitol per day, corresponding to 1000 ml of B. Braun Mannitol 20% Solution for Infusion can be administered if considered necessary by the prescribing physician.

- Prevention of acute renal failure (after positive response to test infusion):

As a rule, 1 – 1.5 g of mannitol per kg body weight (BW), are infused over 1.5 – 4 hours, corresponding to approx. 5 – 7.5 ml/kg BW of B. Braun Mannitol 20% Solution for Infusion.
Usually the dose is adjusted so as to achieve a diuresis of not less than 30 – 50 ml/hour.

Test infusion

In patients with marked oliguria or those suspected to have insufficient renal function, intravenous infusion of a test dose of about 0.2 g of mannitol per kg BW, corresponding to 1 ml of B. Braun Mannitol 20% Solution for Infusion, within 3 – 5 minutes, should lead to a diuresis of not less than 30 – 50 ml per hour during the following 2 – 3 hours. If adequate diuresis is not achieved, a second test infusion may be administered. If diuresis is still insufficient, the further infusion of mannitol solutions is contraindicated.

- Reduction of intracranial pressure:

As a rule, 1.5 – 2 g of mannitol per kg BW, corresponding to approx. 7.5 – 10 ml/kg BW of B. Braun Mannitol 20% Solution for Infusion are infused over 30 – 60 minutes.

If rapid reduction of intracranial pressure is to be achieved, 1 – 1.5 g of mannitol per kg BW, corresponding to approx. 5 – 7.5 ml/kg BW of B. Braun Mannitol 20% Solution for Infusion may be infused over 10 minutes.

When used preoperatively, the dose should be administered 1 to 1.5 hours before surgery to obtain the maximum effect.

In low-weight and debilitated patients a dose of 0.5 g/kg BW, corresponding to approx. 2.5 ml/kg BW of B. Braun Mannitol 20% Solution for Infusion may be sufficient.

The usual interval between individual infusions is 4 – 6 hours. Attention must be paid to the serum osmolality which must not exceed 320 mOsm/l. Adequate reduction of the intracranial pressure is not achievable with higher serum osmolality values but the incidence of adverse effects is markedly increased.

- Forced diuresis to promote elimination of renally excreted toxic substances:

Initially, 25 g of mannitol are given, corresponding to approx. 125 ml of B. Braun Mannitol 20% Solution for Infusion followed by a dose yielding a diuresis of not less than 100 – 150 ml/hour, preferably, however, about 500 ml/hour, maintaining a positive fluid balance of 1 – 2 litres.

- Supportive systemic therapy of acute glaucoma:

As a rule, approx. 1.5 g of mannitol per kg BW, corresponding to approx. 7.5 ml/kg BW of B. Braun Mannitol 20% Solution for Infusion are infused over 30 – 60 minutes.

Paediatric patients

Mannitol dosage requirements for patients up to 12 years have not been definitely established. In the literature, the same test dose as for adults is recommended and therapeutic doses between 0.25 g – 2 g of mannitol/kg BW are considered adequate, i.e. 1.25 – 10 ml of B. Braun Mannitol 20% Solution for Infusion.

Method of administration

Intravenous use.

The solutions must be infused strictly intravenously, otherwise, in cases of paravenous infusion the solutions may cause tissue irritation. In the case of the 15 % and 20 % solutions, even tissue necroses and/or compartment syndrome may occur due to their high osmolarity.

As a rule, the solution is administered as short-term infusion.

When administering B. Braun Mannitol 20% Solution for Infusion an administration set with a filter should be used for infusion.

4.3 Contraindications

Mannitol solutions must not be given in cases of

- Hypersensitivity to mannitol
- Persistent oligo- or anuria after test infusion;
- Severe cardiac insufficiency;
- Pulmonary oedema;
- Dehydration;
- Hyperosmolality of the serum, i. e. > 320 mOsm/kg,
- Intracranial bleeding;
- Obstructions in the urinary tract.

4.4 Special Warnings and Precautions for Use

Special warnings

This solution is only indicated for osmotherapy.

The solution should be administered with caution in cases of hypervolaemia.

In cases of oligo- or anuria, osmotherapy with mannitol solutions should only be performed after a successful test infusion.

The solution should be administered with great caution in cases of compensated cardiac insufficiency, since rapid expansion of extracellular space may cause cardiac failure.

Precautions for use

The patient's cardiovascular status should be carefully assessed before starting osmotherapy and should be monitored during therapy.

Sufficient hydration of the patient should be ensured before beginning of osmotic diuresis. Dehydration should therefore be corrected before start of therapy.

If irregularities of renal function appear during osmotherapy that may indicate reversible vacuolisation, treatment should be stopped immediately in order to prevent progress to irreversible nephrosis.

Clinical monitoring during osmotherapy should include checks of water, electrolyte and acid-base balance, serum osmolarity, renal function, heart function and blood pressure.

The efficacy of all osmotherapeutic agents decreases after repeat therapy, since the osmotic gradient decreases e.g. between blood and eye or brain tissue.

As mannitol slowly enters the brain tissue rebound intracranial hypertension may occur especially after repeated doses.

For monitoring of the urine excretion use of a closed collecting system is recommended.

Mannitol solutions must not be infused through the same infusion line simultaneously with, before, or after the transfusion of blood because of the danger of pseudo-agglutination.

Interference with laboratory tests:

Mannitol interferes with the determination of

- inorganic phosphate in blood, resulting in too high or too low values and
- ethylene glycol.

4.5 Interaction with Other Medicinal Products and Other Forms of Interaction

Ciclosporine

In patients having undergone kidney transplantation and receiving medication with ciclosporines, vacuolisation of the kidney has been observed after administration of mannitol solutions.

Diuretics

The dose of the solution must be carefully adjusted in patients on concomitant medication with other diuretics.

Cardiac glycosides

Because osmotherapy may lead to increased potassium losses, the serum potassium level must be carefully monitored in patients on digitalis therapy. This is even more important if these patients are also receiving diuretics.

On the other hand digitalis clearance may be enhanced under osmotherapy with mannitol, so also the digitalis dosage must be carefully controlled.

Lithium

Mannitol increases the renal excretion of lithium. Therefore the concentration of lithium in serum should be monitored in patients receiving lithium containing medicaments.

4.6 Fertility, pregnancy and lactation

Pregnancy

Mannitol passes the placental barrier.
There are no or limited data from the use of Mannitol in pregnant women Animal studies do not indicate direct or indirect harmful effects with respect to pregnancy, embryonal/fetal development, parturition or postnatal development, and clinical reports on such effects are not known so far.
Yet caution should be exercised when administering mannitol solutions to pregnant women and the doses should be chosen as low as possible.

Breast-feeding

It is not known whether mannitol/metabolites are secreted into breast milk. Therefore the solutions should be administered to breast-feeding women with due caution.

4.7 Effects on Ability to Drive and Use Machines

This medicinal product has no influence on the ability to drive and use machines.

4.8 Undesirable Effects

Most of the undesirable effects listed below are dose-dependent and may therefore be considered symptoms of intoxication, see also **4.9.** Frequencies stated below are estimated with reference to usual clinical doses.

Mainly, depending on the dose and the patient's clinical condition, electrolyte and fluid imbalances with hyper- or hyponatraemia, hyper- or hypokalaemia, and hyper- or dehydration may occur (details see below).

Definition of frequency terms used in this section
Rare (≥1/10,000 to < 1/1,000 of treated patients)
Very rare (< 1/10,000 of treated patients)

Immune system disorders

Very rare : Hypersensitivity reactions, either local reactions such as rhinitis, urticaria, rash, or systemic anaphylactic reactions like fever, oedema, respiratory distress, hypotension, tachycardia or anaphylactic shock.

Metabolism and nutrition disorders

Rare : Acidosis, hyponatraemia, hyperkalaemia, hyperhydration, in particular at the beginning of osmotherapy with mannitol infusions and in cases of overdose.

Very rare Hypernatraemia, hypokalaemia, dehydration due to polyuria occurring at a later stage after longer lasting administration of mannitol solution.

Nervous system disorders

Rare : Headache, dizziness, convulsions.

Eye disorders

Rare : Blurred vision.

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Cardiac disorders
Rare : Acute volume overload of the cardiovascular system (especially after too rapid infusion or overdose of mannitol and in particular under the conditions of insufficient urine production).

Gastrointestinal disorders
Rare : Dryness of mouth, nausea, vomiting, upper abdominal trouble.

Musculoskeletal and connective tissue disorders
Rare : Transient muscle rigidity.

Renal and urinary disorders
Rare : Urine retention, polyuria that rapidly converts to oliguria, uricosuria

General disorders and administration site conditions
Rare : Chills, fever, arm pain, backache, angina-like chest pain, thirst Irritation of the veins and phlebitis after infusion into small veins, skin necrosis

4.9 Overdose

Symptoms
Too rapid infusion or overdose of mannitol can cause acute volume overload of the cardiovascular system, in particular under the conditions of insufficient urine production.

Excessive osmotherapy may lead to major fluid and electrolyte losses leading to circulatory disorders.

In patients with healthy kidneys intoxications with mannitol have been observed with dosages of 400 – 1200 g given over 2 days.

Severe intoxications have been observed in patients with renal insufficiency with dosages of 100 – 400 g given over 1 – 3 days.

As a result of the osmotic effect of mannitol solutions intoxication may manifest as acute fluid overload, electrolyte imbalances and central nervous disorders. Clinical manifestations of intoxication include hypertension, congestive heart failure, pulmonary and peripheral oedema, states of confusion, lethargy, stupor, sometimes coma, convulsions, nausea, vomiting and polyuria, possibly followed by oligo- or anuria, as a consequence of fluid loss hypotension and tachycardia.

Large doses of mannitol in the presence of acidosis may cause toxic damage of the central nervous system.

High doses of mannitol may provoke acute renal failure. There are findings indicating that simultaneous administration of diuretics or pre-existent renal impairment may increase the likelihood of acute renal failure to occur. Beginning renal failure can be reversed by haemodialysis.

Treatment
Immediate stop of infusion, correction of electrolyte and water imbalances, haemodialysis for protection of renal function and elimination of mannitol in order to reduce serum osmolarity, in particular in cases of oligo- or anuria.

Fluid and electrolyte losses are to be corrected by infusion of electrolyte solutions. Specific electrolyte substitution may be performed by means of electrolyte concentrates that may be added to the infusion solution. Electrolyte solutions should be administered separately from the mannitol solutions, e.g. through a multitube catheter.

5 PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic Properties

Pharmacotherapeutic group
Solutions producing osmotic diuresis, mannitol,
ATC Code: B05BC01

Mechanism of action, therapeutic effect
The therapeutically active ingredient in B. Braun Mannitol 20% Solution for Infusion is the polyol mannitol (molecular weight: 182 Daltons).

Due to its osmotic effect it causes a shift of fluid from the intracellular to the extracellular space.

Effect in kidneys
Mannitol passes into the renal tubules by glomerular filtration and only a minor proportion is re-absorbed in the tubules. In the tubules it exerts its osmodiuretic effect by increase of the osmotic pressure resulting in blockage of the absorption of water from the glomerular filtrate. Mannitol infusions lead to an increase of the renal perfusion and, consequently, to an increase of the glomerular filtration rate.

On account of its positive effect on urine production and excretion, mannitol can prevent functional renal failure and transition to organic renal damage, respectively.

Mannitol increases the excretion of sodium which is accompanied by a relatively greater excretion of water; the excretion of potassium, chloride, calcium, phosphate, lithium, magnesium, urea, and uric acid is increased, too.

Effect on the brain
Mannitol does not pass across the blood-brain barrier except when there are very high mannitol plasma levels or in acidosis. As long as the blood-brain barrier function is intact, an osmotic gradient between blood and brain tissue is built up, and fluid is withdrawn from the brain tissue. Thus an existing brain oedema is reduced and the intracranial pressure decreases.

If, however, the blood-brain barrier function is impaired, e.g. in acidosis, mannitol slowly passes into the brain tissue and consecutively the osmotic gradient may be inverted. This results in a fluid shift into brain tissue and eventually the intracranial pressure may increase again (rebound effect).

Effect on the eyes
Mannitol reduces intra-ocular pressure by building up an osmotic gradient. Thus fluid is withdrawn from the anterior eye chamber and the intra-ocular pressure is reduced.

5.2 Pharmacokinetic Properties

Absorption
As the solution is administered intravenously the bioavailability is 100 per cent.

Distribution
After intravenous infusion mannitol is distributed exclusively in the extracellular space, i.e. in plasma and interstitial space.

Osmotic diuresis begins about 1 – 3 hours after infusion of mannitol solution. Reduction of intracranial pressure is to be expected 15 min. after infusion. The effect lasts for about 3 – 8 hours. Decrease of intraocular pressure begins after about 30 – 60 min after infusion and the effect lasts for about 4 – 6 hours.

Biotransformation
Mannitol is not metabolised to a significant degree. Only a very minor proportion is utilised in the liver for glycogen synthesis.

Elimination
In the kidneys mannitol undergoes glomerular filtration and only 10 per cent are re-absorbed to the circulation in the tubules. The elimination half-life is approx. 100 min. It is longer (up to 36 hours) in renal insufficiency. Under the conditions of haemodialysis the half-life time is 6 hours, in patients on peritoneal dialysis up to 21 hours.

Linearity/Non-linearity
Repeated administration leads to a decrease of the intensity and the duration of the effect of mannitol solutions (tachyphylaxis).

5.3 Preclinical Safety Data
Non-clinical data reveal no special hazard for humans based on conventional studies of safety pharmacology, repeated dose toxicity, genotoxicity, carcinogenic potential, toxicity to reproduction and development.

6 PHARMACEUTICAL PARTICULARS

6.1 List of Excipients:
Water for injections

6.2 Incompatibilities
As a matter of principle, solutions for osmotherapy should not be mixed with other medicaments on account of their concentration of active substance and the particulars of clinical use.

The addition of potassium or sodium chloride to B. Braun Mannitol 20% Solution for Infusion may cause precipitation of mannitol. Therefore electrolyte solutions should be administered separately, e.g. through a multitube catheter.

6.3 Shelf Life
- unopened:
Polyethylene containers: 3 years

– after first opening the container
Not applicable. Administration should commence immediately after connecting container to the giving set. Used containers must not be stored for re-use.

– after dilution or reconstitution according to directions
Not applicable

6.4 Special Precautions for Storage
Do not store above 30°C.

6.5 Nature and Contents of Container
- Bottles of low density Polyethylene (LDPE), containing 250 ml, 500 ml,
  available in packs of 30 x 250 ml, 10 x 500 ml.

Not all pack sizes may be marketed.

6.6 Instructions for Disposal and other Handling
No special requirements for disposal.

The product should be stored between 20° and 30°C as exposure to lower temperatures may cause the deposition of crystals.

The solution is supplied in single-use containers. Unused contents of a container once opened must be discarded.

The solution should only be administered if it is clear, colourless and if the container and its closure are undamaged.

7 REVISION DATE
06.2021

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Product registration holder
and manufactured by:
B. Braun Medical Industries S/B
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